Essential Quantum Instability of the Perturbative Yang-Mills Vacuum.

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(ricevuto il 24 Settembre 1986)

Summary. — We prove that there exist gauge field configurations whose energy density is lower than that of the perturbative Yang-Mills vacuum by a quantity $\Delta E \sim b\Lambda^4$ (Λ is the ultraviolet cut-off) where b remains finite when $g \to 0$. This striking physical situation, which prevents the use of perturbation theory even at extremely short distances, is what has been called *essential instability*.

PACS. 11.10. – Field theory.

1. – Introduction.

The question of the stability of the classical ground state (zero field) of the non-Abelian Yang-Mills theory under quantum fluctuations has attracted the theorists' attention since a long time. To our knowledge SAVVIDY (1) was the first to point out in 1977, within the one-loop approximation to the effective potential of a pure SU_2 Yang-Mills theory, that due to quantum fluctuations the classical ground state (*i.e.* the perturbative ground state) does not correspond to a local minimum of the energy density, and a local minimum was found for constant (chromo-)magnetic field H^* (Savvidy's state)

$$gH^* \simeq \Lambda^2 \exp\left[-\frac{24\pi^2}{11g^2(\Lambda)}\right], \tag{1.1}$$

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(1) G. K. SAVVIDY: *Phys. Lett. B*, **71**, 133 (1977).

Λ^2 being the ultraviolet cut-off. The interest of (1.1) lies in its perfect consistency with the renormalization group requirement; indeed, solving (1.1) for $g^2(\Lambda)$, one gets the well-known asymptotic freedom (AF) expression of $g^2(\Lambda)$.

Subsequently, Savvidy's calculation was found to contain a basic flaw (²), in that the state he found was itself unstable due to a peculiar feature of the dynamics of gauge fields in an external constant magnetic field. As shown by LANDAU in the early 30's the energy spectrum of a charged relativistic particle in a costant magnetic field H in the z-direction is given by

$$E_n(p_z) = [p_z^2 + gH(2n+1) - 2gHS_z]^{\frac{1}{2}}\,. \tag{1.2}$$

For $|S|\geqslant 1$, (1.2) implies the existence of a portion of the spectrum ($n=0$, $S_z=1$, $|p_z|<(gH)^{\frac{1}{2}}$) with imaginary energy, thus signalling a basic instability.

A considerable amount of work has been devoted to try and understand the effect of such an instability on the perturbative vacuum, and its possible role in the fundamental property of color confinement. However, only recently (³) it has been possible to go effectively beyond a somewhat heuristic stage, by use of a variational approach based on Gaussian wave functionals, endowed with perturbative (in g) gauge invariance. As we shall describe in sect. **2**, the first results that were obtained have been rather surprising, showing the existence of a Savvidy's state where the energy density has a local minimum for a value of the magnetic field

$$gH_V^* = \Lambda^2 \exp\left[-12\pi^2/11g^2(\Lambda)\right], \tag{1.3}$$

which does not agree with the AF prediction (1.1). As a consequence the long-distance properties of the theory, that should be responsible for colour confinement, cannot be in agreement with perturbation theory (PT). For, if AF were right, by inserting the AF prediction of $g^2(\Lambda)$ in (1.3) one would obtain a divergent ($\sim\Lambda$) magnetic field and an equally divergent ($\sim\Lambda^2$) energy density gap with respect to the perturbative ground state. Divergences that on the real ground state can only be more severe, due to the variational nature of our calculation.

The calculation of ref. (³) did not, however, exhaust the possibilities of the variational techniques developed and a fully consistent calculation which takes into account all terms to $O(g^2)$ was still lacking. In particular, the very important point of whether the interaction between the unstable modes and the stable modes (for which E_n in (1.2) is real) could change our conclusions was only argued in the negative, but not conclusively established.

(²) N. K. NIELSEN and P. OLESEN: *Nucl. Phys. B*, **134**, 376 (1978).
(³) M. CONSOLI and G. PREPARATA: *Phys. Lett. B*, **154**, 411 (1985).

This paper intends to fill the gaps left by the previous work in two different directions: to give a full two-loop analysis of the variational calculation, and to establish the « nonrenormalization » of the nonperturbative effects due to the unstable modes. In the course of our two-loop analysis a new term in the energy density was found whose effect is even more striking than (1.3). I shall indeed prove that the Savvidy's state with minimum energy must be associated with a magnetic field gH^*_G which is *proportional* to Λ^2, the constant of proportionality being finite when $g^2 \to 0$.

Thus, there exists no choice of g^2 such that for scales smaller than a finite scale the perturbative ground state is a good representation of the real ground state. I have called this disquieting phenomenon the *essential quantum instability* of the perturbative ground state. I shall briefly comment about the drastic consequences of this result on the prevailing picture of Yang-Mills theories at the end of this paper.

Let me finally mention that this approach has been recently criticized (4). It is hoped that this paper will give convincing evidence of the unfoundedness of those criticisms.

2. – The original variational calculation.

In ref. (3) a variational calculation was presented of the energy density of the trial state (I adopt here the same notations of that paper)

$$\psi[A^\alpha] = G[\eta^\alpha]\Gamma[\eta^\alpha]\,, \tag{2.1}$$

where the gauge field in the timelike gauge $\boldsymbol{A}^\alpha(\boldsymbol{x})$ ($\alpha = 1\ 2,\ 3,\ SU_2$ index) is written as

$$\boldsymbol{A}^\alpha(\boldsymbol{x}) = \boldsymbol{f}^\alpha(\boldsymbol{x}) + \boldsymbol{\eta}^\alpha(\boldsymbol{x})\,,$$

and $\boldsymbol{f}^\alpha$ is chosen to represent a chromomagnetic field along the z-direction and in the 3-direction of isospace, *i.e.* modulo a gauge transformation one has

$$f^\alpha_k(\boldsymbol{x}) = \delta_{k2}\delta^{\alpha 3}Hx_1\,. \tag{2.3}$$

In eq. (2.1) $G[\eta^\alpha]$ is a Gaussian wave functional

$$G[\eta^\alpha] = (\det G)^{-\frac{1}{4}} \exp\left[-\tfrac{1}{4}\int \mathrm{d}\boldsymbol{x}\,\mathrm{d}\boldsymbol{y}\,\eta^\alpha_i(\boldsymbol{x})\,G^{-1}{}^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y})\,\eta^\beta_j(\boldsymbol{y})\right], \tag{2.4}$$

(4) L. Maiani, G. Martinelli, G. Rossi and M. Testa: *A constant chromomagnetic field leads nowhere*, Rome preprint (November 1985).

where the inverse propagator $G^{-1\alpha\beta}_{ij}(\boldsymbol{x}, \boldsymbol{y})$ contains an infinite number of variational parameters (see eq. (2.13)). The functional $\Gamma[\eta^\alpha]$ is a finite-degree polynomial in the fields η, chosen in such a way as to implement the approximate requirement of gauge invariance:

$$\{D_k^{\alpha\beta}(f) + g\varepsilon^{\alpha\beta\gamma}\eta_k^\gamma(x)\}\frac{\delta}{\delta\eta_k^\beta(x)}\,\psi[\eta] = 0\,, \tag{2.5}$$

where the covariant derivative $D_k^{\alpha\beta}(f)$ is given by

$$D_k^{\alpha\beta}(f) = \delta^{\alpha\beta}\,\partial_k + g\varepsilon^{\alpha\beta\gamma} f_k^\gamma(x)\,. \tag{2.6}$$

Recalling that in the present Schrödinger formalism the electric field $E_k^\alpha(x) = -\, i\delta/\delta\eta_k^\alpha(x)$, we immediately recognize in (2.5) the Gauss law.

In order to evaluate the energy density of the state (2.1), one first writes down the energy density as

$$\mathscr{H}(\boldsymbol{x}) = \tfrac{1}{2}\sum_\alpha[\boldsymbol{E}^{2\alpha}(\boldsymbol{x}) + \boldsymbol{B}^{2\alpha}(\boldsymbol{x})]\,, \tag{2.7}$$

where

$$B_k^\alpha(\boldsymbol{x}) = \varepsilon_{ijk}\big(\partial_j A_k^\alpha(x) + g/2\varepsilon^{\alpha\beta\gamma}A_j^\beta(x)A_k^\gamma(x)\big)\,, \tag{2.8}$$

and then one computes the expectation value of the operator (2.7), on the state (2.1). However, here we encounter a first problem for, as is well known, after having fixed the timelike gauge we have still at our disposal a time-independent gauge transformation. This circumstance makes it impossible to define on the space of gauge-invariant states a scalar product that is normalizable. The way out of this difficulty is well known ([5]); fix a three-dimensional gauge (which for convenience we shall take $D_k^{\alpha\beta}(f)\eta_k^\beta(\boldsymbol{x}) = 0$) and write the scalar product as

$$\langle\psi|\varphi\rangle = \int[\mathrm{d}\eta]\prod_{\boldsymbol{x}}\delta(\mathrm{D}\eta)\,\Delta[\eta]\,\psi^*[\eta]\varphi[\eta]\,, \tag{2.9}$$

where $\Delta[\eta]$ is the Faddeev-Popov determinant, associated with our gauge-fixing, which is evaluated in appendix A. It turns out, however, that the Faddeev-Popov determinat plays no role in our calculation. The last ingredient we need in order to complete the calculation of ref. ([3]) is a technique to solve the contraints imposed by the Gauss law (2.5). From the detailed analysis reported in appendix B, we learn that we can represent the logarithm of the

([5]) L. D. Faddeev and V. N. Popov: *Phys. Lett. B*, **25**, 29 (1967).

functional $\Gamma[\eta]$ as a power series in the coupling constant g:

$$\ln \Gamma[\eta] = \sum_{k=1}^{\infty} g^k W^{(k+2)}[\eta] \,, \tag{2.10}$$

where the functionals $W^{(k+2)}$ are monomials in the fluctuating field $\eta_i^\alpha(x)$ of degree $k+2$, whose explicit form can be obtained from a set of recursive equations. In ref. (3) only the lowest-order contribution, $O(g^2)$, to the energy density has been considered, implying that on our wave functional the Gauss law is satisfied to the same order.

From the results of appendix B one readily derives for the energy density $E(H)$ the following expression (V is the volume of the system):

$$E(H) = 1/V \frac{\langle\psi|\int d^3x\, \mathscr{H}(\boldsymbol{x})|\psi\rangle}{\langle\psi|\psi\rangle} = \tag{2.11}$$

$$= H^2/2 + 1/V\int d^3\boldsymbol{x} \left\{\frac{1}{8}\, G_{ii}^{-1\alpha\alpha}(\boldsymbol{x},\boldsymbol{x}) + \frac{1}{2}\, O_{ij}^{\alpha\beta}(\boldsymbol{x})\, G_{ji}^{\beta\alpha}(\boldsymbol{x},\boldsymbol{x}) + \right.$$

$$+ g^2/4\, \varepsilon^{\alpha\beta\gamma}\varepsilon^{\alpha\beta'\gamma'}[G_{jk}^{\beta\gamma}(\boldsymbol{x},\boldsymbol{x})\, G_{jk}^{\beta'\gamma'}(\boldsymbol{x},\boldsymbol{x}) + G_{jj}^{\beta\beta'}(\boldsymbol{x},\boldsymbol{x})\, G_{kk}^{\gamma\gamma'}(\boldsymbol{x},\boldsymbol{x}) + G_{jk}^{\beta\gamma'}(\boldsymbol{x},\boldsymbol{x})\, G_{jk}^{\beta'\gamma}(\boldsymbol{x},\boldsymbol{x})] -$$

$$\left. - g^2/8\, \varepsilon^{\alpha\beta\gamma}\varepsilon^{\alpha'\beta'\gamma'}\int d^3y (-D^{-2})^{\alpha\alpha'}(\boldsymbol{x},\boldsymbol{y})[G_{jk}^{\beta\beta'}(\boldsymbol{x},\boldsymbol{y})\, G^{-1}{}_{jk}^{\gamma\gamma'}(\boldsymbol{x},\boldsymbol{y}) - \delta_{jk}^{\beta\beta'}(\boldsymbol{x},\boldsymbol{y})\, \delta_{jk}^{\gamma\gamma'}(\boldsymbol{x},\boldsymbol{y})]\right\},$$

where the operator $O_{ij}^{\alpha\beta}(x)$ is given by

$$D_{ij}^{\alpha\beta}(\boldsymbol{x}) = \delta^{\alpha 3}\delta^{\beta 3}\delta_{ij}(-\Delta_{\boldsymbol{x}}) + (\delta^{\alpha\beta} - \delta^{\alpha 3}\delta^{\beta 3})[-\Delta_{\boldsymbol{x}} + 2ix_1\partial_2 + g^2H^2x_1^2] + + 2gH\varepsilon^{\alpha\beta 3}\varepsilon^{ij3}\,, \tag{2.12}$$

and the « propagator » G is represented as

$$G_{ij}^{\alpha\beta}(\boldsymbol{x},\boldsymbol{y}) = \sum_N \frac{1}{2\lambda_N}\, \psi_{iN}^\alpha(\boldsymbol{x})\, \psi_{jN}^{\beta *}(\boldsymbol{y}) \tag{2.13}$$

with $\{\psi_N^\alpha\}$ the complete set of eigenfunctions of the operator (2.12), with eigenvalues E_N^2 (3), obeying the « transversality » condition

$$D_j^{\alpha\beta}(f)\,\psi^\beta{}_{jN}(\boldsymbol{x}) = 0\,, \tag{2.14}$$

and λ_N the parameters of the variational calculation.

Equation (2.11) differs from the analogous eqs. (14)-(19) of ref. (3) in the coefficient of the last term, which is now a factor of two smaller. The problem of this factor of two (*) is worth considering in some detail. Let us take for simplicity the case $H = 0$.

(*) This argument is due to M. CONSOLI.

We can parametrize the propagator as

$$G_{ij}^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y}) = \delta^{\alpha\beta}\int \frac{\mathrm{d}^3 p}{(2\pi)^3\, 2\lambda(p)}\left(\delta_{ij} - \frac{p_i p_j}{p^2}\right)\exp[i\boldsymbol{p}(\boldsymbol{x}-\boldsymbol{y})] \tag{2.15}$$

and eq. (2.11) becomes

$$E(H=0) = 3/2\int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}\left[\lambda(p) + \frac{p^2 + M^2(p)}{\lambda(p)}\right] + \lambda\text{-independent terms}, \tag{2.16}$$

where

$$M^2(p) = g^2/6\int \frac{\mathrm{d}^3\boldsymbol{p}'}{(2\pi)^3\lambda(\boldsymbol{p}')}\left\{4 - 3\left(1 + \frac{(\boldsymbol{p}-\boldsymbol{p}')^2}{p^2 p'^2}\right)\frac{\lambda^2(p')}{(\boldsymbol{p}-\boldsymbol{p}')^2}\right\}. \tag{2.17}$$

One easily checks that the function $M^2(p)$ is free of quadratic divergences provided the eigenvalues $\lambda(p')$ in eq. (2.17) are replaced by their lowest-order expression $\lambda(p) = |\boldsymbol{p}|$. This is a direct consequence of our solution of the Ward identities up to order g^2. If we are interested, however, in a variational procedure and, therefore, in fully minimizing eq. (2.16) with respect to $\lambda(p)$, we discover that $\lambda(p) = |\boldsymbol{p}|(1 + O(g^2))$ is not a solution due to the asymmetric structure $(\boldsymbol{p} \leftrightarrow \boldsymbol{p}')$ of the second term in eq. (2.17). As a consequence, if we separate out in eq. (2.16) the low-frequency modes, the minimization equations with respect to their eigenvalues would still contain quadratic divergences if the ultraviolet modes are treated in perturbation theory. The unfortunate mistake (the factor of two) done in ref. (3) when solving the Ward identities was difficult to detect due to the peculiar fact that, in this case, the energy density of the U-modes was free of quadratic divergences and their wave functions got renormalized in a completely consistent way.

In the next section I shall discuss a way to improve the variational ansatz, which involves the full structure of the Yang-Mills Hamiltonian and, in particular, the cubic term

$$H_3 = g/2\varepsilon^{\alpha\beta\gamma}\int \mathrm{d}^3x(D_j^{\alpha\mu}\eta_k^\mu - D_k^{\alpha\mu}\eta_j^\mu)\eta_j^\beta(x)\eta_k^\gamma(x), \tag{2.18}$$

which so far has played no role in the variational calculations. We shall see that our way of proceeding has a precise parallel in the strategy followed by perturbation theory.

But before describing the improved variational ansatz let us briefly discuss the results of ref. (3).

As mentioned in the introduction, the striking property of the operator (2.12) appearing in the quadratic part of the Hamiltonian is that its spectrum *is not positive semi-definite*. There are, in fact, eigenmodes of $O_{ij}^{\alpha\beta}(\boldsymbol{x})$, the U-modes, that are associated to negative eigenvalues. They correspond to Landau states with $n = 0$, $S_3 = +1$ and momentum along the third direction $|P_3| < |gH|^{\frac{1}{2}}$. For such modes any perturbative analysis is obviously bound to fail.

Neglecting for a moment all other modes, that are called stable modes or S-modes, the contribution $E_U(H)$ of the U-modes to the energy density can be readily computed from (2.11) with the result

$$E_U(H) = -H^2/2(u' - u^2/2)\,, \tag{2.19}$$

where

$$u' = g^2/4\pi \int_{-1}^{+1} \mathrm{d}x \left(\xi(x) - \frac{1-x^2}{\xi(x)}\right), \tag{2.20}$$

$$u = g^2/4\pi \int_{-1}^{+1} \frac{\mathrm{d}x}{\xi(x)} \tag{2.21}$$

and

$$\xi(x) = (gH)^{-\frac{1}{2}} \lambda_0[(gH)^{\frac{1}{2}} x]\,, \tag{2.22}$$

$\lambda_0(p_3)$ being the variational parameters associated with the unstable modes. By minimizing (2.19) with respect to $\xi(x)$, one obtains a minimum E_U^*

$$E_U^*(H) \simeq H^2/4\,, \tag{2.23}$$

for

$$\lambda_0^*(p_3) \simeq \left[p_3^2 + 4gH \exp\left[-\frac{4\pi^2}{g^2}\, u^*\right]\right]^{\frac{1}{2}} \tag{2.24}$$

and

$$u^* = 1\,. \tag{2.24'}$$

The remarkable aspect of (2.23) is its independence from g, indicating the essentially classical nature of the unstable modes, which are thus able to screen half of the classical energy density $H^2/2$ of the background field (2.3).

Turning now to the S-modes, ignoring again their coupling to the U-modes, their contribution $E_S(H)$ to the energy density is well known [1,2] and is given by

$$E_S(H) = E(0) - \frac{11}{48\pi^2}\, g^2 H^2 \ln\left(\frac{\Lambda^2}{gH}\right) + O\left(g^4 H^2 \ln \frac{\Lambda^2}{gH}\right). \tag{2.25}$$

Thus, modulo the interaction between S- and U-modes, we can write for the total contribution of (2.11)

$$E(H) \simeq E(0) + \frac{H^2}{4} - \frac{11}{48\pi^2} g^2 H^2 \ln\left(\frac{\Lambda^2}{gH}\right), \tag{2.26}$$

which differs from the calculations based on the effective potential in one fundamental respect: the classical term is only half of what it should be. The disastrous consequences of this result on asymptotic freedom and perturbative QCD have been amply discussed [6], and here I need not delve into this any longer.

In order to maintain our confidence in (2.26), however, we must ascertain whether the neglected interaction between S- and U-modes might modify the logarithmic term, thus changing the conclusions based on the assumed validity of (2.26).

Little thought is needed to convince ourselves that it would be physically very strange for a « good » theory, such as a Yang-Mills gauge theory, to generate an extra logarithmic contribution to (2.26) from the coupling of S- and U-modes. Indeed such a circumstance would mean that the basically infra-red phenomenon (see (2.21)) associated with the U-modes, that must disappear when an infra-red cut-off $M_0 \gg (gH)^{\frac{1}{2}}$ is introduced, can be sensitive to a very fine granularity of space-time (the ultraviolet cut-off Λ) (*).

Nevertheless it would certainly be nice to see this fact explicitly arising from a direct computation. In ref. [3], it was suggested that the neglected terms have the only effect to renormalize multiplicatively the u-parameter appearing in (2.19) without changing conclusions (2.20) and (2.21). A general proof of this fact shall be presented in sect. **4**, here we wish only to remark that our result hinges heavily upon the nonperturbative minimization procedure that has been applied to the energies of both the U- and the S-modes. In this sense, contrary to a recent criticism [4], our variational calculation *does* go definitely beyond a perturbative analysis.

3. – The improved variational calculation: the essential quantum instability.

I shall now describe an improvement of the original variational calculation, which closely parallels the strategy followed in perturbation theory.

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[6] G. PREPARATA: in *Proceedings of the International Europhysics Conference on High Energy Physics, Bari, 1985*, edited by L. NITTI and G. PREPARATA (Bari, 1986), p. 130.

(*) Apparently these are the simple considerations that subtend the statement in T. D. LEE: *Particle Physics and Introduction to Field Theory* (Harwood Academic Publishers, 1984), p. 126; that the energy of a quantum soliton must be cut-off independent. In the line of thought of ref. [2] we may look at the U-modes just as a quantum soliton (see appendix E).

This improvement is aimed principally at keeping under control the ultraviolet behavior of the approximation. This shall be accomplished by introducing a mass scale E_0 $\left(E_0 \gg (gH)^{\frac{1}{2}}\right)$ in such a way that the S-modes for which $E_N \geqslant E_0$ shall be treated perturbatively, while those with $E_N < E_0$ are analysed variationally. Thus we shall decompose the fluctuating field $\eta_i^\alpha(\boldsymbol{x})$ as

$$\eta_i^\alpha(\boldsymbol{x}) = h_i^\alpha(\boldsymbol{x}) + l_i^\alpha(\boldsymbol{x})\,, \tag{3.1}$$

where $h_i^\alpha(\boldsymbol{x})$ contains the modes with $E_N > E_0$—the h-modes—, while $l_i^\alpha(\boldsymbol{x})$ is built up by the S-modes with $E_N < E_0$ and by the U-modes, which will be globally called l-modes.

The functional $\Gamma[\eta]$ appearing in (2.1) will be improved as follows:

$$\Gamma[\eta] = \left(1 + g\delta^{(3)}[\eta]\right) \exp\left[gW^{(3)}[\eta]\right], \tag{3.2}$$

where the functional $\delta^{(3)}[\eta]$, differently from $W^{(3)}[\eta]$, contains transverse fields only. $\delta^{(3)}[\eta]$ is further decomposed as

$$\delta^{(3)}[\eta] = \delta_1^{(3)}[\eta] + \delta_2^{(3)}[\eta]\,, \tag{3.3}$$

where $\delta_1^{(3)}$ contains one l-mode and two h-modes, and $\delta_2^{(3)}$ contains two l-modes and one h-mode.

Let us begin with $\delta_1^{(3)}[\eta]$. We write

$$\delta_1^{(3)} = \frac{1}{\sqrt{2!}} \int\limits_{\boldsymbol{x}_1\boldsymbol{x}_2\boldsymbol{x}} A_{i_1 i_2 i}^{\alpha_1\alpha_2\alpha}(\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x})\, a_{i_1}^{\alpha_1}(\boldsymbol{x}_1)^\dagger\, a_{i_2}^{\alpha_2}(\boldsymbol{x}_2)^\dagger\, b_i^\alpha(\boldsymbol{x})^\dagger, \tag{3.4}$$

where one has introduced the creation operators for the h-modes

$$a_i^\alpha(\boldsymbol{x})^\dagger = 1/2\, h_i^\alpha(\boldsymbol{x}) - \int\limits_{\boldsymbol{y}} H_{ij}^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y})\, \frac{\delta}{\delta h_j^\beta(\boldsymbol{x})}\,, \tag{3.5}$$

and for the l-modes,

$$b_i^\alpha(\boldsymbol{x})^\dagger = 1/2\, l_i^\alpha(\boldsymbol{x}) - \int\limits_{\boldsymbol{y}} L_{ij}^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y})\, \frac{\delta}{\delta l_j^\beta(\boldsymbol{x})}\,, \tag{3.6}$$

and the propagator is decomposed as

$$G_{ij}^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y}) = L_{ij}^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y}) + H_{ij}^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y})\,. \tag{3.7}$$

Up to $O(g^2)$ the matrix elements we must consider are those of the quadratic (H_2) and cubic (H_3) parts of the Hamiltonian, which we compute in appendix C.

Note that in our calculation only the quadratic Hamiltonian $H_2^{(h)}$ of the h-modes is needed, for the contribution of $H_2^{(l)}$ (*) can be absorbed in a redefinition of the propagator L, which is to be determined variationally.

One notes immediately that the amplitudes $A^{\alpha_1\alpha_2\alpha}_{i_1i_2i}$ appear quadratically in the matrix element of H_2 but only linearly in H_3.

By minimizing with respect to such amplitudes, we get (eq. (C.6))

$$A^{\alpha_1\alpha_2\alpha}_{i_1i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}) = -1/4\int_{\boldsymbol{y}_1\boldsymbol{y}_2\boldsymbol{y}} K^{(\beta_1\beta_2)\beta}_{(j_1j_2)j}(\boldsymbol{y}_1\boldsymbol{y}_2\boldsymbol{y})\cdot (\delta)^{\beta_1\alpha_1}_{j_1i_1}(\boldsymbol{y}_1,\boldsymbol{x}_1)(H^{-1})^{\beta_2\alpha_2}_{j_2i_2}(\boldsymbol{y}_1,\boldsymbol{x}_2)(L^{-1})^{\beta\alpha}_{ji}(\boldsymbol{x},\boldsymbol{y}) \tag{3.8}$$

with

$$K^{(\alpha_1\alpha_2)\alpha}_{(i_1i_2)i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}) = 1/2[K^{\alpha_1\alpha_2\alpha}_{i_1i_2i}(\boldsymbol{x}_1\boldsymbol{x}_2,\boldsymbol{x}) + (1\leftrightarrow 2)] \tag{3.9}$$

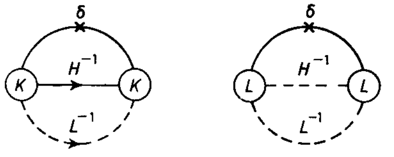
and

$$K^{\alpha_1\alpha_2\alpha}_{i_1i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}) = \sqrt{2!}\,\varepsilon^{\lambda\mu\nu}\int_{\boldsymbol{z}}\{D^{\lambda\varrho}_{[j}(\boldsymbol{z})L^{\varrho\,\alpha}_{k]i}(\boldsymbol{z},\boldsymbol{x})H^{\mu\alpha_1}_{ji_1}(\boldsymbol{z},\boldsymbol{x}_1)H^{\nu\alpha_2}_{ki_2}(\boldsymbol{z},\boldsymbol{x}_2) + D^{\lambda\varrho}_{[j}H^{\varrho\,\alpha_1}_{k]i_1}(\boldsymbol{z},\boldsymbol{x}_1)\big(H^{\mu\alpha_2}_{ji_2}(\boldsymbol{z},\boldsymbol{x}_2)L^{\nu\alpha}_{ki}(\boldsymbol{z},\boldsymbol{x}) + H^{\nu\alpha_2}_{ki_2}(\boldsymbol{z},\boldsymbol{x}_2)L^{\mu\alpha}_{ji}(\boldsymbol{z},\boldsymbol{x})\big)\,. \tag{3.10}$$

The contribution $E_1^{(3)}$ to the energy density is

$$E_1^{(3)} = -g^2/4V\int_{\substack{\boldsymbol{x}_1\boldsymbol{x}_2\boldsymbol{x}\\ \boldsymbol{y}_1\boldsymbol{y}_2\boldsymbol{y}}} K^{(\alpha_1\alpha_2)\alpha}_{i_1i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x})(\delta)^{\alpha_2\beta_2}_{i_1j_1}(\boldsymbol{x}_1,\boldsymbol{y}_1)H^{-1\,\alpha_2\beta_2}_{i_2j_2}(\boldsymbol{x}_2,\boldsymbol{y}_2)\cdot L^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y})K^{(\beta_1\beta_2)\beta}_{(j_1j_2)j}(\boldsymbol{y}_1,\boldsymbol{y}_2,\boldsymbol{y})\,. \tag{3.11}$$

The structure of $E_1^{(3)}$ is worth noting. Indeed we can associate to it the diagram reported in fig. 1a), where K is the two h- and one l-modes' unamputated vertex and the two-body operator δ represents the insertion of the quadratic Hamiltonian $H_2^{(h)}$. The interest of this observation lies in the not



a) b)

Fig. 1. – The diagrammatic interpretation of a) (3.11), and b) (3.13). Continuous lines represent h-modes, while dashed lines represent l-modes.

(*) According to the discussion above, the high-frequency part $H^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y})$ is given by its lowest-order expression $H^{\alpha\beta}_{ij} = \sum_{E_N\geqslant E_0} 1/2E_N\,\psi_N\psi_N^*$.

too surprising fact that the solution of our variational problem in the presence of a small parameter (g) exhibits a perturbative structure (*).

As for $\delta_2^{(3)}$, the wave functional containing two l-modes and one h-mode, one writes

$$\delta_2^{(3)} = 1/\sqrt{2!}\int\limits_{x_1x_2x} B^{(\alpha_1\alpha_2)\alpha}_{(i_1i_2)i}(x_1, x_2, x)\, b^{\alpha_1}_{i_1}(x_1)^\dagger\, b^{\alpha_2}_{i_2}(x_2)^\dagger\, a^{\alpha}_{i}(x)^\dagger . \tag{3.12}$$

Proceeding exactly as before for the contribution $E_2^{(3)}$ to the energy density, we obtain

$$E_2^{(3)} = -\, g^2/4V \int\limits_{\substack{x_1x_2x\\ y_1y_2y}} L^{(\alpha_1\alpha_2)\alpha}_{(i_1i_2)i}(x_1, x_2, x)(L^{-1})^{\alpha_1\beta_1}_{i_1j_1}(x_1, y_1)(L^{-1})^{\alpha_2\beta_2}_{i_2j_2}(x_2, y_2)\cdot \cdot\,\delta^{\alpha\beta}_{ij}(x, y)\, L^{(\beta_1\beta_2)\beta}_{(j_1j_2)j}(x_1 y_2, y)\,, \tag{3.13}$$

where for $L^{(\alpha_1\alpha_2)\alpha}_{(i_1i_2)i}(x_1, x_2, x)$ one derives an expression identical to (3.9), and (3.10) with $L^{\alpha\beta}_{ij}$ and $H^{\alpha\beta}_{ij}$ interchanged. The diagram associated with (3.13) is depicted in fig. 1b).

At this point all we need to do in order to complete our calculation is to evaluate $E_1^{(3)}$ and $E_2^{(3)}$, a task that requires a substantial amount of labor, which can luckily be saved, as we shall demonstrate in the next section.

Two crucial features of these contributions, however, can be established without too much effort. The first has to do with the question of the cancellation of the quadratic divergences contained in (2.11). A straightforward calculation (appendix D) shows that the quadratic divergences contained in $E_1^{(3)}$ are precisely right to compensate those appearing in (2.11), thus implementing a basic requirement of full fledged gauge invariance.

The other crucial feature is the presence in $E_2^{(3)}$ of a term (see appendix D)

$$-\frac{H^2u^2}{8} \tag{3.14}$$

that cancels half of the positive and « stabilizing » contribution of the U-modes to the quartic part of the Hamiltonian.

As a result the full contribution E_U to the energy density of the U-modes, replacing (2.16), is

$$E_U = -\frac{H^2}{2}\left(u - \frac{u^2}{4}\right), \tag{3.15}$$

(*) Incidentally our construction shows that the Rayleigh-Ritz approximation in the Schrödinger picture can be systematically improved by means of a perturbationlike expansion [7].

[7] J. N. Cornwall, R. Jackiw and E. T. Tomboulis: *Phys. Rev. D*, **1**, 2428 (1974).

whose minimum occurs at $u^* = 2$ and equals

$$E_{U\,\min} = -\frac{H^2}{2}, \tag{3.16}$$

thus completely cancelling the classical energy density! Neglecting the interaction between S-modes and U-modes, as it shall be fully justified in the next section, the energy density of our trial state is

$$E(H) = E(0) - \frac{11 g^2 H^2}{48\pi^2} \ln\left(\frac{\Lambda^2}{gH}\right) + O(g^2 H^2). \tag{3.17}$$

This is the central result of this paper and the basis of the contention that the perturbative ground state of a Yang-Mills theory is *essentially unstable.* Even though we have not proved that our state is the real ground state, we have nevertheless demonstrated that there exists a perfectly well-defined and respectable configuration of the gauge fields whose energy density is infinitely (when $\Lambda \to \infty$) lower than that of the perturbative ground state. Indeed the disappearance from (3.17) of the classical term implies that the minimum occurs at $H^* \sim \Lambda^2$ and the energy difference at the minimum is given by $E(0) - E(H^*) \sim \Lambda^4$. Given this very extreme situation, in no circumstance can the perturbative ground state be a good approximation to the real ground state, for the real ground state must be lower in energy than our state, which is already infinitely (when $\Lambda \to \infty$) distant from the perturbative one. We hope that this brief discussion provides a sufficiently clear illustration of the rather unusual notion of essential instability. We shall, however, resume this discussion in the final section.

4. – The essential instability survives renormalization.

Let us first summarize the consequences of the improved variational ansatz that we have just described:

i) The contribution to the energy density of the U-modes takes the form

$$E_U = E_U^{(1)} + E_U^{(2)} \simeq -H^2/2(u - u^2/4) \tag{4.1}$$

corresponding to the diagrams in fig. 2.

ii) All quadratic divergencies to $O(g^2)$ cancel.

iii) The coupling of the U-modes to the S-modes induces logarithmic corrections $O(g^2 \ln \Lambda^2/gH)$ to both the u- and the u^2-term, whose coefficient can be computed directly with a considerable amount of labor.

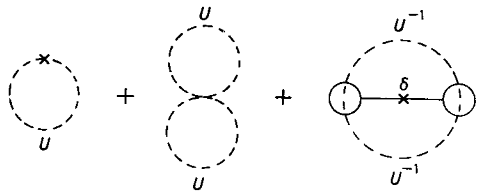


Fig. 2. – Diagrams contributing to E_U (eq. (4.1)). The dashed lines correspond to U-modes.

iv) While for the logarithmic corrections to the u-term our ansatz exhausts them, for the u^2-term the situation is different. One gets in fact only the part originating from the (nonperturbative) minimization in the λ_N's of the S-modes.

As stressed in ref. ([3]) there is another part whose origin lies beyond our approximation, involving the consideration of 4th-order monomials in both transverse and longitudinal field modes.

The issue we want to address ourselves to in this section is thus to find out whether the logarithmic corrections to (4.1), induced by the interaction among U- and S-modes, could modify the conclusions reached in the preceding section about the essential instability of the perturbative Yang-Mills vacuum. As already noted, there is a direct route to answer this crucial question, which is barred by considerable, if not unsurmountable, difficulties.

However, as we shall now see, all such labor can be saved at once by making use of a general argument, which will demonstrate that the renormalization of (4.1) does not alter our conclusions.

In order to accomplish our task, we shall carry out an analysis of the coupling between S- and U-modes that is perturbative in character (*).

That this way of proceeding is a sensible one has already been indicated by the previous variational calculation, where we have noted the close parallelism with perturbation theory, which shows that the problem of the renormalization of the U-modes by the S-modes is perturbative (in the S-modes) in character. Perturbative character which obviously is not shared by the whole problem studied in this paper.

Thus separating the S- from the U-modes

$$A_\mu^\alpha(x) = \eta_{\mu S}^\alpha(x) + \eta_{\mu U}^\alpha(x) + f_\mu^\alpha(x)\,, \tag{4.2}$$

(*) We show in appendix F that in a simplified example ($\lambda\varphi^4$ with negative renormalized mass squared) the result we are about to prove holds in the variational framework of sect. **3**.

and fixing the « background gauge »

$$D_\mu^{\alpha\lambda}(f+\eta_U)\,\eta_{\mu S}^{\lambda}(x)=0\,, \tag{4.3}$$

we define the S-modes, partition function Z

$$Z_S=\int[\mathrm{d}\eta_S]\prod_x\delta\big(D(f+\eta_U)\eta_S\big)\,\Delta[\eta_S]\exp[iS']\,, \tag{4.4}$$

where the action is defined as ($S(A)$ being the Yang-Mills action)

$$S'=S(A)-S_{\text{eff}}\,, \tag{4.5}$$

$$S_{\text{eff}}=S(f+\eta_U+\bar{\eta}_S) \tag{4.6}$$

and (*)

$$\bar{\eta}_{\mu S}^{\alpha}(x)=-\frac{1}{2}\int(O^{-1})_{\mu\nu}^{\alpha\beta}(x,y)\,\frac{\delta^3 S}{\delta\eta_\nu^\beta(x)\,\delta\eta_{\mu_1}^{\alpha_1}(x_1)\,\delta\eta_{\mu_2}^{\alpha_2}(x_2)}\,\eta_{\mu_1 U}^{\alpha_1}(x_1)\,\eta_{\mu_2 U}^{\beta_2}(x_2)\,; \tag{4.7}$$

O^{-1} is the inverse of the operator defined in (2.12).

Finally $\Delta[\eta_S]$ in (4.4) is the Faddeev-Popov determinant, appropriate to the « background gauge » (4.3).

Our problem is then to compute the renormalization of the effective Hamiltonian operator in the space of U-modes, $H_{\text{eff}}^{(u)}$. It is clear that calling $\tilde{\theta}_{\mu\nu}(x)$ the stress-energy tensor relative to the action

$$\tilde{S}=S(A)-S(f+\eta_U)+S(f+\eta_U+\bar{\eta}_S)\,, \tag{4.8}$$

one has

$$H_{\text{eff}}^{(u)}=1/Z_S\int[\mathrm{d}\eta_S]\prod_x\delta(D\eta_S]\,\Delta[\eta_S]\int\mathrm{d}^3x\,\tilde{\theta}_{00}(\boldsymbol{x},0)\exp[iS']\,,$$

eqs. (4.8) generate an infinite set of Feynman diagrams. However, for our purpose—the calculation of $O(g^2\ln\Lambda^2/gH)$ corrections to (4.1)—we need only

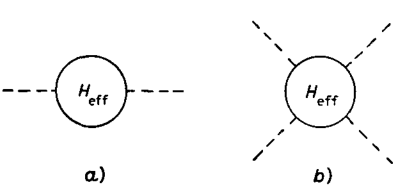


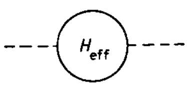
Fig. 3. – Coefficient of *a*) the quadratic and *b*) quartic parts of $H_{\text{eff}}^{(u)}$ (eq. (4.6)).

(*) The field $\bar{\eta}_S$ is precisely what reproduces, once inserted in the action S, the third diagram of fig. 2.

consider the one-loop contributions to the coefficients of the quadratic and quartic (in η_U) parts of $H^{(u)}_{\text{eff}}$ (see fig. 3).

Beside the operator $H^{(u)}_{\text{eff}}$ it is useful to consider the analogous coefficients of the full Hamiltonian operator $H^{(u)}$, defined by changing in (4.4) S' with $S(A)$.

Let us consider the coefficient of the quadratic part first (fig. 3*a*)). To one loop it coincides with the coefficient of $H^{(u)}$. In view of the conservation properties of $\tilde{\theta}_{00}(x)$ and of the gauge we work in (*), we can write the diagrammatic equation

(4.9)  $$= \frac{1}{Z_B}\int d^3x\,\tilde{\theta}^{(2)}_{00}(f+\eta_\mu)\,,$$

where $\tilde{\theta}^{(2)}_{\mu\nu}(A)$ is the quadratic part of the stress-energy tensor and to $O(g^2 \ln \Lambda^2/gH)$ we have ([8])

$$1/Z_B = 1 - 11/24\pi^2 g^2 \ln(\Lambda^2/gH)\,. \tag{4.10}$$

Equations (4.5) and (4.18) imply that the renormalization of the quadratic part of the effective Hamiltonian can be effectively compensated by a corresponding renormalization of the wave function of the U-mode. Indeed computing now the expectation value of $H^{(u)}_{\text{eff}}$ on the U-modes Gaussian state, we have for the renormalized u-term $E^{(1)}_{UR}$:

$$E^{(1)}_{UR} = -H^2/2u_R = -H^2/2u(1 - 11/24\pi^2 g^2 \ln(\Lambda^2/gH))\,. \tag{4.11}$$

Let us now turn to the coefficient in fig. 3*b*). Again its renormalization properties shall be inferred from those of the full Hamiltonian operator $H^{(u)}$. It is only necessary to draw all the diagrams contributing to both coefficients up to one loop to realize at once that the two classes coincide, but for the set reported in fig. 4, which represents the insertion of the stress-energy tensor on the external U-modes. Indeed these diagrams contribute to the full Hamiltonian $H^{(u)}$ but not to $H^{(u)}_{\text{eff}}$.

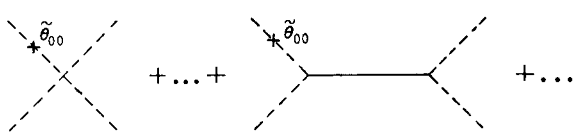


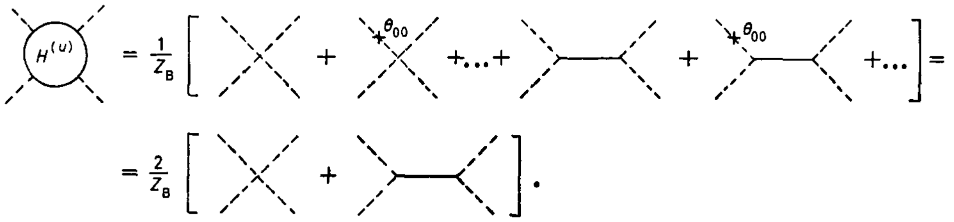
Fig. 4. – Diagrams contributing to $H^{(u)}$ but not to $H^{(u)}_{\text{eff}}$; the dashed lines represent the U-modes, while the continuous lines represent the S-modes.

(*) In the background gauge there is only *one* renormalization constant (see, for instance, ref. ([8])).

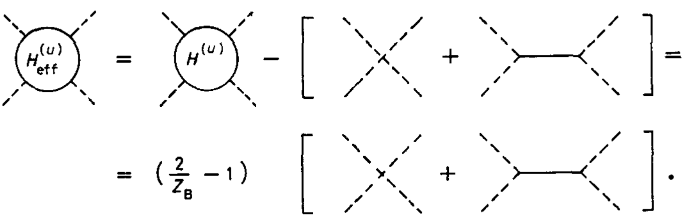
([8]) L. F. Abbott: *Nucl. Phys. B*, **185**, 189 (1981).

A straightforward calculation shows that the diagrams in fig. 4 give precisely the same contribution as those without insertions. At this point our task is almost completed, for we know that the coefficient of the quartic part of $H^{(u)}$ is also multiplicatively renormalized by $1/Z_B$ (eq. (4.6)) (*).

Thus we many write diagrammatically (to one loop)

(4.12) 

Thus we have

(4.13) 

Taking the expectation value of (4.11) on the Gaussian state for U-modes, we obtain for the renormalized u-term $E^{(2)}_{UR}$

(4.14)
$$E^{(2)}_{UR} = -\frac{H^2}{2}\frac{u^2}{4}\left(\frac{2}{Z_B}-1\right) = -\frac{H^2}{2}\frac{u^2}{4}\left(1-2\frac{11}{24\pi^2}g^2\ln\frac{\Lambda^2}{gH}\right) \simeq$$
$$\simeq -\frac{H^2}{2}\frac{u^2}{4}\left(1-\frac{11}{24\pi^2}g^2\ln\frac{\Lambda^2}{gH}\right)^2 .$$

For the complete renormalized contribution we obtain

(4.15)
$$E_{UR} = E^{(1)}_{UR} + E^{(2)}_{UR} = -H^2/2(u_R - u_R^2/4),$$

where

(4.16)
$$u_R = u(1-11/24\pi^2 g^2 \ln(\Lambda^2/gH)).$$

By comparing (4.15) with (4.1) we see that the only effect of renormalization is to change u to u_R (eq. (4.14)), which has obviously no consequence upon the minimization of the energy density.

(*) This is a consequence of the renormalizability of the Yang-Mills theory.

Thus the essential instability of the perturbative vacuum induced by the nonperturbative U-modes' amplitudes *is not removed* by the renormalization due to the coupling of the U-modes to the S-modes.

One should notice that (4.16), apart from the coefficients of the u_R^2-term and of $\ln(\Lambda^2/gH)$ in (4.16), is totally consistent with the conclusions of ref. (3). Furthermore, the origin of the different factors appearing in (4.12) and (4.13) can be traced to the circumstance that the renormalization of the u^2-term receives contributions from both the magnetic and the electric parts of the Hamiltonian, while the lowest-order contribution originates from the magnetic term only.

We conclude this section by recalling that the « nonrenormalization » result that we have just obtained is physically quite transparent and, so to say, inevitable. Had we indeed found a logarithmic renormalization of the lowest-order contribution of the U-modes to the energy density $(-H^2/2)$, this would have meant that, upon measuring the energy density of field modes that are completely insensitive to the ultraviolet cut-off, such as the U-modes, one could nevertheless discover the existence of the cut-off.

5. – Conclusions.

The original aim of this paper was to fill the gaps left by the discussion in ref. (3), by means of an improved variational ansatz, which could simulate some of the important features of the perturbative coupling between S- and U-modes. In the course of the analysis described in sect. **3** we have discovered the existence of the term (3.14), which makes the conclusions of ref. (3) even more drastic. Indeed our central result (eq. (3.17)) for the difference between the energy density of our state and the perturbative vacuum:

$$E(H) - E(0) = -11/48\pi^2 g^2 H^2 \ln(\Lambda^2/gH) + O(g^2 H^2) \tag{5.1}$$

implies a local minimum

$$gH^* = a\Lambda^2 \tag{5.2}$$

and the energy difference at the minimum

$$E(H^*) - E(0) = -b\Lambda^4, \tag{5.3}$$

where a and b $(a, b > 0)$ remain finite when $g \to 0$.

What does all this mean? The answer is very simple: there exist Savvidy states of the gauge field, characterized by a chromomagnetic field H^* in a fixed direction in both space and colour space, whose energy density is, according to (5.1) and (5.3), infinitely lower (when $\Lambda \to \infty$) than the perturbative ground state.

And, if we still believe that QCD is the best candidate for a theory of the strong interactions, we have no other choice but to *abandon* altogether perturbation theory and look, may be along the directions open by this work, for the real QCD ground state. For one thing we know from the variational nature of our calculation: the energy of the real ground state must be equal or lower than (5.1). We have also seen that, due to the peculiar dynamical behaviour of the U-modes, there is a definite advantage to « switch on » in the classical vacuum a constant chromomagnetic field H^*. This makes it plausible that the real QCD ground state should be not too far from the one we have discussed here.

Finally a brief comment on the results of sect. **4**. According to them (and the discussion in appendix F) we have now a full proof that the coupling of the U-modes to the S-modes does not change in any fundamental way the « classical » dynamics of the U-modes. This fact, whose physical soundness as well as theoretical origin were clearly spelled out in ref. (3), has been the subject of a rather pointed controversy (4). In ref. (4) it has been suggested that the renormalization of the U-modes induced by the S-modes should tame the instabilities caused by the U-modes, and reinstate the validity of perturbation theory at short distances. Not only does this circumstance seem hardly believable, but according to our proof it is just incorrect.

* * *

I wish to thank most warmly M. CONSOLI for collaboration in the initial stage of this work and for many enlightening discussions. I also wish to thank S. TAZZARI, Director of Laboratori Nazionali di Frascati of INFN, for his kind hospitality at Frascati, where part of this work was carried out.

APPENDIX A

We shall now give a derivation of eq. (2.9) together with an explicit expression for the Faddeev-Popov determinant $\Delta[\eta]$, associated to the gauge fixing

$$D_j^{\alpha\beta}(\boldsymbol{x})\,\eta_j^\beta(\boldsymbol{x}) = 0\,. \tag{A.1}$$

Following FADDEEV-POPOV (5), the problem we wish to solve is to extract from the matrix elements of any gauge-invariant operator on gauge-invariant wave functionals a universal (infinite) constant, that may therefore be ignored.

Let us define the gauge-invariant functional

$$\Delta_f[\eta]^{-1} = \int [\mathrm{d}\Omega(\boldsymbol{x})] \prod_{\boldsymbol{x}} \delta[f(\eta^{\Omega})] , \tag{A.2}$$

where $f(\eta) = 0$ is a generic gauge fixing, $\Omega(\boldsymbol{x})$ belongs to the class of time-independent gauge transformations, and

$$\eta_k(\boldsymbol{x})^{\Omega} = \Omega(\boldsymbol{x})\, \eta_k(\boldsymbol{x})\, \Omega^{-1} + 1/g \partial_k \Omega(\boldsymbol{x})\, \Omega^{-1}(\boldsymbol{x}) \tag{A.3}$$

is the fluctuation gauge transformed by $\Omega(x)$.

The most general matrix element

$$\langle \psi | O\left(\eta, -i\frac{\delta}{\delta\eta}\right) | \varphi \rangle = \int [\mathrm{d}\eta]\, \psi^*[\eta]\, O\left[\eta, -i\frac{\delta}{\delta\eta}\right] \varphi[\eta] \tag{A.4}$$

according to (A.2) can be written as

$$\langle \psi | O\left(\eta, -i\frac{\delta}{\delta\eta}\right) | \varphi \rangle = \int [\mathrm{d}\Omega(\boldsymbol{x})] \prod_{\boldsymbol{x}} \delta[f(\eta^{\Omega})]\, [\mathrm{d}\eta^{\Omega}]\, \Delta_f'[\eta] \cdot \\ \cdot \psi[\eta]^* O\left[\eta, -i\frac{\delta}{\delta\eta}\right] \varphi[\eta] , \tag{A.5}$$

which, using the gauge invariance of $\Delta_f[\eta]$, $\varphi[\eta]$, $\psi[\eta]$ and $O[\eta, -i(\delta/\delta\eta)]$, can be cast in the form

$$\langle \psi | O\left(\eta, -i\frac{\delta}{\delta\eta}\right) | \varphi \rangle = \int [\mathrm{d}\Omega(\boldsymbol{x})] \prod_{\boldsymbol{x}} \delta(f(\eta^{\Omega}))\, [\mathrm{d}\eta^{\Omega}]\, \Delta_f[\eta^{\Omega}]\, \psi[\eta^{\Omega}]^* \cdot \\ \cdot O\left(\eta^{\Omega}, -i\frac{\delta}{\delta\eta^{\Omega}}\right) \varphi^*[\eta^{\Omega}] = \int [\mathrm{d}\Omega(\boldsymbol{x})] \int [\mathrm{d}\eta] \prod_{\boldsymbol{x}} \delta(f(\eta))\, \Delta_f[\eta]\, \psi[\eta]^* O\left(\eta, -i\frac{\delta}{\delta\eta}\right) \varphi[\eta] = \\ = V_G \int [\mathrm{d}\eta] \prod_{\boldsymbol{x}} \delta(f(\eta))\, \Delta_f[\eta]\, \psi[\eta]^* O\left(\eta, -i\frac{\delta}{\delta\eta}\right) \varphi[\eta] . \tag{A.6}$$

The above equation solves our problem, due to the constancy of $V_G = \int [\mathrm{d}\Omega(x)]$.

From (A.2) we see that in order to evaluate $\Delta_f[\eta]$ we need only integrate upon infinitesimal transformations $\Omega(\boldsymbol{x}) = (1 + \omega(\boldsymbol{x}))$.

Thus we have

$$\Delta[\eta]^{-1} = \int [\mathrm{d}\Omega]\, \delta(f(\eta^{\Omega})) = \int [\mathrm{d}\omega] \prod_{\boldsymbol{x}} \delta\left[f(\eta) + \frac{\delta f}{\delta\omega}\,\omega\right] = \\ = \int [\mathrm{d}\omega] \prod_{\boldsymbol{x}} \delta\left(\frac{\delta f}{\delta\omega}\,\omega\right) = \det\left(\frac{\delta f}{\delta\omega}\right)^{-1} , \tag{A.7}$$

i.e.

$$\Delta_f[\eta] = \det\left(\frac{\delta f}{\delta\omega}\right) . \tag{A.8}$$

Setting now

$$f(\eta) = D_i^{\alpha\beta}(f)\,\eta_j^{\beta}(\boldsymbol{x})\,,$$

and recalling that

$$\frac{\delta\eta_i^{\alpha}(\boldsymbol{x})}{\delta\omega^{\beta}(\boldsymbol{y})} = [D_i^{\alpha\beta}(f) + g\varepsilon^{\alpha\beta\gamma}\eta_i^{\gamma}(\boldsymbol{x})]\,\delta(\boldsymbol{x}-\boldsymbol{y})\,, \tag{A.9}$$

we finally obtain for $\Delta[\eta]$, appearing in (2.9), the expression

$$\Delta[\eta] = \det\{[D_j^{\alpha\gamma}(f)\,D_j^{\gamma\alpha'}(f) + g\varepsilon^{\alpha\beta\gamma}D_j^{\beta\alpha'}(f)\,\eta_j^{\gamma}(\boldsymbol{x})]\,\delta^3(\boldsymbol{x}-\boldsymbol{y})\}\,. \tag{A.10}$$

Introducing, as usual, Grassmann variables $c^{\alpha}(\boldsymbol{x})$ and $\bar{c}^{\beta}(\boldsymbol{x})$ we may write (neglecting irrelevant constant factors)

$$\Delta[\eta] = \int[\mathrm{d}c][\mathrm{d}\bar{c}]\exp\left[-\left\{\int\mathrm{d}^3x\,\bar{c}(\boldsymbol{x})(D^2)\,c(\boldsymbol{x}) + g\int\mathrm{d}^3x\,\mathrm{d}^3y\,\bar{c}(\boldsymbol{x})\,\Delta(\boldsymbol{x},\boldsymbol{y})\,c(\boldsymbol{y})\right\}\right], \tag{A.11}$$

where

$$\Delta^{\alpha\beta}(\boldsymbol{x},\boldsymbol{y}) = g\varepsilon^{\lambda\beta\nu}D_j^{\alpha\lambda}(\boldsymbol{x})\,\eta_j^{\nu}(\boldsymbol{x}) \tag{A.12}$$

and

$$(D^2)^{\alpha\beta}(\boldsymbol{x},\boldsymbol{y}) = D_j^{\alpha\lambda}(\boldsymbol{x})\,D_j^{\lambda\beta}(\boldsymbol{x})\,\delta^3(\boldsymbol{x}-\boldsymbol{y})\,. \tag{A.13}$$

A simple calculation shows that to $O(g^2)$ no ultraviolet divergences arise proportional to the unstable-mode amplitude, thus justifying our neglect of $\Delta[\eta]$ both in ref. (3) and throughout this paper.

Appendix B

The Gauss law constraint that we wish to solve is

$$\left\{D_k^{\alpha\beta}(\boldsymbol{x})\,\frac{\delta}{\delta\eta_k^{\beta}(\boldsymbol{x})} + g\varepsilon^{\alpha\varrho\beta}\eta_k^{\varrho}(\boldsymbol{x})\,\frac{\delta}{\delta\eta_k^{\beta}(\boldsymbol{x})}\right\}\psi[\eta] = 0\,. \tag{B.1}$$

Setting

$$\psi[\eta] = G[\eta]\,\Gamma[\eta] \tag{B.2}$$

and

$$\Gamma[\eta] = \exp\,[W[\eta]]\,, \tag{B.3}$$

(B.1) becomes

$$\begin{aligned} D_k^{\alpha\beta}(\boldsymbol{x})&\left\{-1/2\int\mathrm{d}^3y\,G_k^{-1}{}^{\beta\lambda}_{l}(\boldsymbol{x},\boldsymbol{y})\,\eta_l^{\lambda}(\boldsymbol{y}) + \frac{\delta W}{\delta\eta_k^{\beta}(\boldsymbol{x})}\right\} + \\ &+ g\varepsilon^{\alpha\varrho\beta}\eta_k^{\varrho}(\boldsymbol{x})\left\{-1/2\int\mathrm{d}^3y\,G_k^{-1}{}^{\beta\lambda}_{l}(\boldsymbol{x},\boldsymbol{y})\,\eta_l^{\lambda}(\boldsymbol{y}) + \frac{\delta W}{\delta\eta_k^{\beta}(\boldsymbol{x})}\right\} = 0\,. \end{aligned} \tag{B.4}$$

Using the «transversality» $D_k^{\alpha\beta}(\boldsymbol{x})\, G_k^{-1\,\beta\lambda}{}_l(\boldsymbol{x}, \boldsymbol{y}) = 0$, we obtain

$$\text{(B.5)}\qquad D_k^{\alpha\beta}(\boldsymbol{x})\,\frac{\delta W}{\delta\eta_k^\beta(\boldsymbol{x})} = -g\varepsilon^{\alpha\varrho\beta}\eta_k^\varrho(\boldsymbol{x})\left\{-\frac{1}{2}\int \mathrm{d}^2 y\; G_k^{-1\,\beta\lambda}{}_l(\boldsymbol{x}, \boldsymbol{y})\,\eta_l^\lambda(\boldsymbol{y}) + \frac{\delta W}{\delta\eta_k^\beta(\boldsymbol{x})}\right\}.$$

Expanding $W[\eta]$ as in (2.10) (B.5) takes the recursive form

$$\text{(B.6)}\qquad D_k^{\alpha\beta}(\boldsymbol{x})\,\frac{\delta W^{(3)}}{\delta\eta_k^\beta(\boldsymbol{x})} = +\frac{1}{2}\,\varepsilon^{\alpha\varrho\beta}\eta_k^\varrho(\boldsymbol{x})\int \mathrm{d}^3 y\; G_k^{-1\,\beta\lambda}{}_l(\boldsymbol{x}, \boldsymbol{y})\,\eta_l^\lambda(\boldsymbol{y})$$

and $(k \geqslant 2)$

$$\text{(B.6}')\qquad D_k^{\alpha\beta}(\boldsymbol{x})\,\frac{\delta W^{(2+k)}}{\delta\eta_k^\beta(\boldsymbol{x})} = -\,\varepsilon^{\alpha\varrho\beta}\eta_k^\varrho(\boldsymbol{x})\,\frac{\delta W^{(1+k)}}{\delta\eta_k^\beta(\boldsymbol{x})}\,,$$

whose solutions can be easily written as

$$\text{(B.7)}\qquad \frac{\delta\eta_j^\alpha(\boldsymbol{x})}{\delta W^{(3)}} = \frac{1}{2}\int D_j^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y})\,\varepsilon^{\beta\varrho\sigma}\eta_k^\varrho(\boldsymbol{y})\,G_k^{-1\,\sigma\lambda}{}_l(\boldsymbol{y}, \boldsymbol{z})\,\eta_l^\lambda(\boldsymbol{z})\,\mathrm{d}y\,\mathrm{d}z$$

and for $k \geqslant 2$

$$\text{(B.7}')\qquad \frac{\delta W^{(k+2)}}{\delta\eta_j^\alpha(\boldsymbol{x})} = -\int D_j^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y})\,\varepsilon^{\beta\varrho\sigma}\eta_k^\varrho(\boldsymbol{y})\,\frac{\delta W^{(k+1)}}{\delta\eta_k^\sigma(\boldsymbol{y})}\,,$$

where

$$\text{(B.8)}\qquad D_j^{\alpha\beta}(\boldsymbol{x}, \boldsymbol{y}) = D_j^{\alpha\lambda}(\boldsymbol{x})\left[\frac{1}{D^2(\boldsymbol{x}, \boldsymbol{y})}\right]^{\lambda\beta},$$

with

$$\text{(B.9)}\qquad (D^2)^{\alpha\beta} = D_k^{\alpha\varepsilon}(\boldsymbol{x})\,D_k^{\varepsilon\beta}(\boldsymbol{x})\,,$$

obeys the equation

$$\text{(B.10)}\qquad D_j^{\alpha\beta}(\boldsymbol{x})\,D_j^{\beta\gamma}(\boldsymbol{x}, \boldsymbol{y}) = \delta^3(\boldsymbol{x} - \boldsymbol{y})\,.$$

For our purposes eqs. (A.7) need no further development, for when the electric part of the Hamiltonian is applied to the state (A.2), one obtains

$$\text{(B.11)}\qquad -\frac{1}{2}\,\frac{\delta^2}{\delta\eta_i^\alpha(\boldsymbol{x})\,\delta\eta_i^\alpha(\boldsymbol{x})}\,\psi[\eta] = \frac{1}{4}\,G^{-1\,\alpha\alpha}_{\;\;ii}(\boldsymbol{x}, \boldsymbol{x}) - \frac{1}{2}\,\frac{\delta^2 W}{\delta\eta_i^\alpha(\boldsymbol{x})\,\delta\eta_i^\alpha(\boldsymbol{x})} -$$
$$-\frac{1}{8}\int \mathrm{d}^3 y\,\mathrm{d}^3 z\; G^{-1\,\alpha\lambda}_{\;\;il}(\boldsymbol{x}, \boldsymbol{y})\,G^{-1\,\alpha\mu}_{\;\;im}(\boldsymbol{x}, \boldsymbol{z})\,\eta_l^\lambda(\boldsymbol{y})\,\eta_m^\mu(\boldsymbol{z}) +$$
$$+\frac{1}{2}\int \mathrm{d}^3 y\,\frac{\delta W}{\delta\eta_i^\alpha(\boldsymbol{x})}\,G^{-1\,\alpha\lambda}_{\;\;il}(\boldsymbol{x}, \boldsymbol{y})\,\eta_l^\lambda(\boldsymbol{y}) - \frac{1}{2}\,\frac{\delta W}{\delta\eta_i^\alpha(\boldsymbol{x})}\,\frac{\delta W}{\delta\eta_i^\alpha(\boldsymbol{x})}\,.$$

Keeping terms up to $O(g^2)$, and carrying out the Gaussian functional integration, we obtain

$$\text{(B.12)}\qquad \frac{1}{2}\langle\psi|\sum_\alpha\int \mathrm{d}^3x\,\boldsymbol{E}^2_\alpha(x)|\psi\rangle = \frac{1}{8}\int \mathrm{d}^3x\,G^{-1}(\boldsymbol{x},\boldsymbol{x}) - \\ -\frac{g}{8}\,\varepsilon^{\beta\varrho\sigma}\varepsilon^{\beta'\varrho'\sigma'}\int \mathrm{d}^3y\,\mathrm{d}^3y'\,\frac{1}{-D^2}(\boldsymbol{y},\boldsymbol{y}')^{\beta\beta'}\cdot \\ \cdot\{G^{\varrho\varrho'}_{kk'}(\boldsymbol{y},\boldsymbol{y}')\,G^{-1}{}^{\sigma\sigma'}_{kk'}(\boldsymbol{y},\boldsymbol{y}') + \delta^{\sigma'\varrho}_{k'k}(\boldsymbol{y}',\boldsymbol{y})\,\delta^{\sigma\varrho'}_{kk'}(\boldsymbol{y},\boldsymbol{y}')\} + \\ + \text{terms independent of the propagators } G.$$

In (B.12) $\delta^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y})$ is the « transverse » δ-function

$$\text{(B.13)}\qquad \delta^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y}) = \sum_N \psi^\alpha_i(\boldsymbol{x})_N\,\psi^\beta_j(\boldsymbol{y})_N\,,$$

where the functions $\psi^\alpha_{iN}(\boldsymbol{x})$ have been defined in (2.13) and (2.14).

Appendix C

We derive here the relevant formulae needed in the discussion in sect. **3.**

2h-modes + *1l-mode*. We evaluate first ($|0\rangle$ denotes the Gaussian wave functional)

$$\text{(C.1)}\qquad E_2 = \frac{g^2}{2!}\int\langle O|a^{\beta_1}_{j_1}(\boldsymbol{y}_1)\,a^{\beta_2}_{j_2}(\boldsymbol{y}_2)\,b^\beta_j(\boldsymbol{y})\,H_2\,a^{\alpha_1}_{i_1}(\boldsymbol{x}_1)^\dagger\,a^{\alpha_2}_{i_2}(\boldsymbol{x}_2)^\dagger\,b^\alpha_i(\boldsymbol{x})^\dagger|O\rangle\cdot \\ \cdot A^{\beta_1\beta_2\beta}_{j_1j_2j}(\boldsymbol{x}_1,\boldsymbol{y}_2,\boldsymbol{y})\,A^{\alpha_1\alpha_2\alpha}_{i_1i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}) = \\ = g^2\int A^{\beta_1\beta_2\beta}_{j_1j_2j}(\boldsymbol{y}_1,\boldsymbol{y}_2,\boldsymbol{y})\,\delta^{\beta_1\alpha_1}_{i_1a_1}(\boldsymbol{x}_1,\boldsymbol{x}_1)\,H^{\beta_2\alpha_2}_{j_2i_1}(\boldsymbol{y}_2,\boldsymbol{x}_2)\,A^{\beta\beta}_{ji}(\boldsymbol{y},\boldsymbol{x})\,,$$

where use has been made of the easily proven results

$$\text{(C.2)}\qquad [H_2, a^\alpha_i(\boldsymbol{x})^\dagger] = \tfrac{1}{2}\int H^{-1}{}^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y})\,a^\beta_j(\boldsymbol{y})$$

and

$$\text{(C.2')}\qquad [a^\alpha_i(\boldsymbol{x}), a^{\dagger\beta}_j(\boldsymbol{y})] = H^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y})\,,$$

etc.

We next compute

$$\text{(C.3)}\qquad E_3 = \frac{g}{\sqrt{2!}}\int\langle O|H_3\,a^{\alpha_1}_{i_1}(\boldsymbol{x}_1)^\dagger\,a^{\alpha_2}_{i_2}(\boldsymbol{x}_2)^\dagger\,b^\alpha_i(\boldsymbol{x})^\dagger|O\rangle\,A^{\alpha_1\alpha_2\alpha}_{i_2i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}) + \\ + \text{h.c.} = g^2\int_{\boldsymbol{x}_1\boldsymbol{x}_2\boldsymbol{x}} K^{\alpha_1\alpha_2\alpha}_{i_1i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x})\,A^{\alpha_1\alpha_2\alpha}_{i_1i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x})\,,$$

where

$$\text{(C.4)}\quad K^{\alpha_1\alpha_2\alpha}_{i_1i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x})=\sqrt{2!}\,\varepsilon^{\lambda\mu\nu}\int\limits_{\boldsymbol{z}}\{D^{\lambda\varrho}_{[j}(\boldsymbol{z})\,G^{\varrho\alpha_1}_{k]i_1}(\boldsymbol{z},\boldsymbol{x}_1)\left(H^{\mu\alpha_2}_{ji_2}(\boldsymbol{z},\boldsymbol{x}_2)\,L^{\nu\alpha}_{ki}(\boldsymbol{z},\boldsymbol{x})\,+\right.$$

$$\left.+\,H^{\nu\alpha_1}_{ki_1}(\boldsymbol{z},\boldsymbol{x}_2)\,L^{\mu\alpha}_{ji}(\boldsymbol{z},\boldsymbol{x})\right)+D^{\lambda\varrho}_{[j}\,L^{\varrho\,\alpha}_{k]i}(\boldsymbol{z},\boldsymbol{x})\,H^{\mu\alpha_1}_{ji_1}(\boldsymbol{z},\boldsymbol{x}_1)\,H^{\nu\alpha_2}_{ki_2}(\boldsymbol{z},\boldsymbol{x}_2)\}\,.$$

Upon imposing the minimization equation

$$\text{(C.5)}\qquad \frac{\delta}{\delta A^{\alpha_1\alpha_2\alpha}_{i_1i_2i}}\,(E_2+E_3)=0\,,$$

one obtains

$$\text{(C.6)}\quad A^{\alpha_1\alpha_2\alpha}_{i_1i_2i}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x})=$$

$$=-\tfrac{1}{2}\int\limits_{\boldsymbol{y}_1,\boldsymbol{y}_2,\boldsymbol{y}}K^{(\beta_1\beta_2)\beta}_{(j_1j_2)j}(\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{y})\,\delta^{\beta_1\alpha_1}_{j_1i_1}(\boldsymbol{y}_1,\boldsymbol{x}_1)\,H^{-1\,\beta_2\alpha_2}_{\;\;j_2i_2}(\boldsymbol{y}_2,\boldsymbol{x}_2)\,L^{-1\,\beta\alpha}_{\;\;ji}(\boldsymbol{y},\boldsymbol{x})\,,$$

where $K^{(\beta_1\beta_2)\beta}_{(j_1j_2)j}$ is the symmetrized kernel (C.4). Introducing (C.6) into E_2+E_3 one readily obtains (3.11).

2l-model + 1h-mode. An identical procedure leads us to (3.13) in the text.

Appendix D

We shall now show that quadratic divergences to $O(g^2)$ are absent from the sum of the contributions (2.11) and (3.11). In calculating the quadratic divergent pieces we shall employ the leading behaviour of the propagator $H^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y})$,

$$\text{(D.1)}\qquad H^{\alpha\beta}_{ij}(\boldsymbol{x},\boldsymbol{y})\to\delta^{\alpha\beta}\int\frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{1}{2|\boldsymbol{p}|}\exp\left[i\boldsymbol{p}(\boldsymbol{x}-\boldsymbol{y})\right]\left(\delta_{ij}-\frac{p_ip_j}{\boldsymbol{p}^2}\right).$$

Inserting (D.1) into eq. (2.11), we get

$$\text{(D.2)}\quad (2.11)\simeq\frac{2}{3}\,g^2\int\frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{1}{|p|}\,\frac{1}{V}\int L^{\alpha\alpha}_{ii}(\boldsymbol{x},\boldsymbol{x})\,\mathrm{d}^3x\,+$$

$$-\,g^2/3\int\frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{1}{|p|}\,\frac{1}{V}\int L^{\alpha\alpha}_{ii}(\boldsymbol{x},\boldsymbol{x})\,\mathrm{d}^3x+\text{logarithmic terms},$$

while from eq. (3.11) we obtain after some simple algebra

$$\text{(D.3)}\quad (3.11)=-\,g^2/3\int\frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{1}{|\boldsymbol{p}|}\,\frac{1}{V}\int L^{\alpha\alpha}_{ii}(\boldsymbol{x},\boldsymbol{x})\,\mathrm{d}^3x+\text{logarithmic terms}.$$

Putting (D.2) and (D.3) together one obtains the announced cancellation.

Next we prove that $E_2^{(3)}$ gives a term $-H^2/2\,u^2/4$ to the energy density. Upon specializing (3.13) to the U-modes, *i.e.* setting $L_{ij}^{\alpha\beta}(\boldsymbol{x},\boldsymbol{y}) = U_{ij}^{\alpha\beta}(\boldsymbol{x},\boldsymbol{y})$, and taking into account that (within irrelevant terms $= O(p_3)$)

$$D_{[j}^{\alpha\lambda}\, U_{kl]}^{\lambda\alpha}(\boldsymbol{x},\boldsymbol{y}) \simeq 0 \tag{D.4}$$

one obtains

$$E_2^{(3)} \simeq -g^2/2\left[\frac{gH}{4\pi^2}\int\limits_{-\sqrt{gH}}^{\sqrt{gH}}\frac{\mathrm{d}p}{2\lambda(p)}\right]^2, \tag{D.5}$$

recalling that $u = (g^2/4\pi^2)\int\limits_{-\sqrt{gH}}^{\sqrt{gH}}(\mathrm{d}p/\lambda(p))$, one rewrites (D.5) as

$$E_2^{(3)} \simeq -\frac{H^2}{2}\frac{u^2}{4}, \tag{D.5$'$}$$

which is the desired result.

Appendix E

In this appendix we shall show that in the (classical) limit $g \to 0$, the U-modes build up a solitonlike solution.

Let us write the gauge field projection on the U-mode

$$\eta_{\pm\,\mu}^{(u)\alpha}(x) = \sum_n \varphi_{\pm;n}(z,t)\,\psi_{\pm;n}(x_\perp)_\mu^\alpha, \tag{E.1}$$

where the spatial part for the $\pm$-charged mode is given by

$$\psi_n(x_\perp) = \frac{N}{L^{\frac{1}{2}}}\exp\left[\pm i\frac{2\pi}{L}ny\right]\exp\left[-\frac{gH}{2}\left(x-\frac{2\pi}{LgH}n\right)^2\right]. \tag{E.2}$$

As we are interested in the limit $g \to 0$, we can choose the following quantization volume (with periodic boundary conditions):

$$|x|,\,|y| \leqslant L/2 \ll \frac{\pi}{(gH)^{\frac{1}{2}}}, \qquad |z| \leqslant L_z/2 \gg \frac{\pi}{(gH)^{\frac{1}{2}}}$$

so that when $g \to 0$ both L and L_z go to ∞. In such a quantization volume the only mode that survives in (E.2) is the one with $n = 0$, and (E.1) becomes

$$\eta_{\pm\,\mu}^{(u)\alpha}(x) \simeq \frac{1}{L}\,\varphi_{\pm}(z,t)\,\varepsilon_{\pm\mu}^{\alpha}. \tag{E.3}$$

A simple calculation gives for the action

$$S \simeq \int \mathrm{d}z\,\mathrm{d}t \left\{ \dot{\varphi}_+(z,t)^2 - \partial_z \varphi_+(z,r)^2 + gH\varphi_+(z,t)^2 + \right. \qquad \text{(E.4)}$$
$$\left. + (+ \to -) - \frac{g^2}{L^2}[\varphi_+(z,t)\varphi_-(z,t)]^2 \right\},$$

where we have chosen the gauge so as to make $\varphi_\pm$ real.

The classical Euler-Lagrange equations are therefore

$$\text{(E.5)} \quad \begin{cases} \ddot{\varphi}_+(z,t) - \partial_z^2\varphi_+(z,t) - gH\varphi_+(z,t) + 2\dfrac{g^2}{L^2}\varphi_+(z,t)\varphi_-(z,t)^2 = 0, \\[2ex] \ddot{\varphi}_-(z,t) - \partial_z^2\varphi_-(z,t) - gH\varphi_-(z,t) + 2\dfrac{g^2}{L^2}\varphi_-(z,t)\varphi_+(z,t)^2 = 0, \end{cases}$$

which admit a solution $\varphi_+ = \varphi_- = \varphi$ such that

$$\text{(E.5}'\text{)} \qquad \ddot{\varphi} - \partial_z^2\varphi - gH\varphi + \frac{2g^2}{L^2}\varphi^3 = 0\,.$$

The solutions of the two-dimensional equation (E.5′) are well known, in particular the static solutions are

$$\text{(E.6)} \qquad \varphi(z,t) = L\left(\frac{H}{2g}\right)^{\frac{1}{2}} \mathrm{tgh}\left[\left(\frac{gH}{2}\right)^{\frac{1}{2}}(z-z_0)\right].$$

We can now compute the energy. One has

$$\text{(E.7)} \quad E = \int \mathrm{d}z \left\{ \dot{\varphi}(z,0)^2 + \partial_z\varphi(z,0)^2 - gH\varphi(z,0)^2 + \frac{g^2}{L^2}\varphi(z,0)^4 \right\} =$$
$$= -\frac{(gH)^2}{4g^2} L^2 \int_{-L_z/2}^{L_z/2} \mathrm{d}z \left(\mathrm{tgh}\sqrt{\frac{gH}{2}}\,z\right)^4 \simeq -\frac{H^2}{4}V\,,$$

which coincides with (2.20).

Our derivation shows that as far as the classical part of the energy density is concerned all the physics does come from the U-sector, even though, strictly speaking, (E.1) are *not* solutions of the classical field equations, as pointed out in appendix (A.3) of ref. [4].

Appendix F

In this appendix we will analyse the origin of the « nonrenormalization theorem », discussed in sect. **4**, in the simple $\lambda\varphi^4$-theory.

In analogy with our problem we shall consider a Lagrangian with negative (renormalized) squared mass:

$$\mathscr{L} = 1/2\,(\partial_\mu\varphi)^2 - U(\varphi) \tag{F.1}$$

with

$$U(\varphi) = +1/2\,\mu_0^2\varphi^2 + \frac{\lambda}{4!}\varphi^4, \tag{F.2}$$

where μ_0^2 is assumed to generate a negative renormalized mass, *i.e.* we set

$$\mu_0^2 = -\mu^2 + \Lambda^2 \sum_n c_n \lambda^n\,, \tag{F.3}$$

where μ^2 is a finite positive number and the coefficients c_n are suitably chosen so as to cancel the quadratic divergences of perturbation theory. Note that in gauge theories such a cancellation holds automatically. Quantizing the theory in the symmetrical phase $\langle\varphi\rangle = 0$, we decompose φ according to

$$\varphi = \varphi_S + \varphi_U\,, \tag{F.4}$$

where $\varphi_U(\varphi_S)$ refer to an eigenmode expansion with momentum $|\boldsymbol{p}| < \mu$ ($|\boldsymbol{p}| > \mu$), likewise for the propagator $G(\boldsymbol{x}, \boldsymbol{y})$ we have

$$G(\boldsymbol{x}, \boldsymbol{y}) = G_U(\boldsymbol{x}, \boldsymbol{y}) + G_S(\boldsymbol{x}, \boldsymbol{y}) \tag{F.5}$$

with

$$G_U(\boldsymbol{x}, \boldsymbol{y}) = \int\limits_{|\boldsymbol{p}|<\mu} \frac{d^3p}{(2\pi)^3}\frac{1}{2\lambda_U(\boldsymbol{p})}\exp[i\boldsymbol{p}(\boldsymbol{x}-\boldsymbol{y})]\,, \tag{F.5$'$}$$

$$G_S(\boldsymbol{x}, \boldsymbol{y}) = \int\limits_{|\boldsymbol{p}|\geqslant\mu} \frac{d^3p}{(2\pi)^3}\frac{1}{2\lambda_S(\boldsymbol{p})}\exp[i\boldsymbol{p}(\boldsymbol{x}-\boldsymbol{y})]\,, \tag{F.5$''$}$$

computing the energy density on the Gaussian functional, we readily obtain

$$\begin{aligned} E = {} & \frac{1}{8}[G_U^{-1}(\boldsymbol{x}, \boldsymbol{x}) + G_S^{-1}(\boldsymbol{x}, \boldsymbol{x})] + \frac{1}{4}O_S[G_s(\boldsymbol{x}, \boldsymbol{x}) + G_U(\boldsymbol{x}, \boldsymbol{x})] + \\ & + \frac{\lambda}{4}G_S(\boldsymbol{x}, \boldsymbol{x})G_U(\boldsymbol{x}, \boldsymbol{x}) + \frac{\lambda}{8}[G_U(\boldsymbol{x}, \boldsymbol{x})G_U(\boldsymbol{x}, \boldsymbol{x}) + G_S(\boldsymbol{x}, \boldsymbol{x})G_S(\boldsymbol{x}, \boldsymbol{x})] \end{aligned} \tag{F.6}$$

with $O_x = (-\boldsymbol{\nabla}^2 + \mu_0^2)$. Setting

$$\mu_0^2 + \frac{\lambda}{4}\int\limits_{|\boldsymbol{p}|\geqslant\mu}\frac{d^3p}{(2\pi)^3}\frac{1}{\lambda_S(\boldsymbol{p})} = -\mu^2 \tag{F.7}$$

and minimizing in λ_S we obtain the contribution of the S-modes

$$\text{(F.8)} \qquad E_S = \frac{1}{2}\int\limits_{|\boldsymbol{p}|\geqslant\mu} \frac{\mathrm{d}^3 p}{(2\pi)^3}\sqrt{\boldsymbol{p}^2-\mu^2+\lambda/2\,u} = \ldots + \\ + \frac{\mu^2}{2}\frac{\lambda}{32\pi^2}\ln\left(\frac{\Lambda^2}{\mu^2}\right) - \frac{\lambda^2}{256\pi^2}u^2\ln\left(\frac{\Lambda^2}{\mu^2}\right),$$

where

$$\text{(F.9)} \qquad u = G_U(\boldsymbol{x},\boldsymbol{x}) = \int\limits_{|\boldsymbol{p}|<\mu}\frac{\mathrm{d}^3 p}{(2\pi)^3}\frac{1}{2\lambda_U(\boldsymbol{p})}\,.$$

Summing to (F.8) the contribution of the U-modes, we obtain

$$\text{(F.10)} \qquad E \simeq -\frac{\mu^2}{2}u\left(1-\frac{\lambda}{32\mu^2}\ln\frac{\Lambda^2}{\mu^2}\right) + \frac{\lambda}{8}u^2\left(1-\frac{\lambda}{32\pi^2}\ln\frac{\Lambda^2}{\mu^2}\right),$$

which shows that the variational Gaussian approximation produces not only terms of $O(\lambda u \ln(\Lambda^2/\mu^2))$ but also terms $O(\lambda^2 u^2 \ln(\Lambda^2/\mu^2))$, which in a perturbative expansion arise only at the 3-loop level. However, the terms we obtain do not exhaust such a contribution. Indeed we can improve our Gaussian state $|G\rangle$ in the way suggested in sect. **3**. We treat the S-modes perturbatively (*i.e.* we set $\lambda_{S'}(\boldsymbol{p}) = \sqrt{\boldsymbol{p}^2-\mu^2}$) and the contribution of the U-modes in the Gaussian approximation (without minimization in the S-modes) is

$$\text{(F.11)} \qquad E_{0U} = -\frac{\mu^2}{2}u\left(1-\frac{\lambda}{32\pi^2}\ln\frac{\Lambda^2}{\mu^2}\right) + \frac{\lambda}{8}u^2\,.$$

We improve the Gaussian ansatz as

$$\text{(F.12)} \qquad |\psi\rangle = (1+\lambda\delta^{(4)})\,G\,,$$

where

$$\text{(F.13)} \qquad \delta^{(4)} = \frac{1}{\sqrt{2!}}\frac{1}{\sqrt{2!}}\int A(\boldsymbol{x}_1,\boldsymbol{x}_2;\boldsymbol{y}_1,\boldsymbol{y}_2)\,s^\dagger(\boldsymbol{x}_1)\,s^\dagger(\boldsymbol{x}_2)\,u^\dagger(\boldsymbol{y}_1)\,u^\dagger(\boldsymbol{y}_2)\,,$$

and $s^\dagger$ and $u^\dagger$ are the creation operators for S- and U-modes, respectively. Calculating the S-mode quadratic Hamiltonian (*) and the full quartic Hamiltonian on (F.12) up to $Q(\lambda^2)$ and minimizing with respect to the amplitudes A, we obtain the extra contribution

$$\text{(F.14)} \qquad E^{(4)} = -\frac{\lambda^2}{4V}\int\limits_{\boldsymbol{xy}} G_S^2(\boldsymbol{x},\boldsymbol{y})\,G_S(\boldsymbol{x},\boldsymbol{y})\,G_U(\boldsymbol{x},\boldsymbol{y})\,G_U(\boldsymbol{x},\boldsymbol{y})\,,$$

(*) Note that we are now treating the S-modes perturbatively, just as in the proof of our nonrenormalization theorem in sect. **4**.

which modifies (F.11) as

$$E_U \simeq -\frac{\mu^2}{2}u\left(1-\frac{\lambda}{32\pi^2}\ln\frac{\Lambda^2}{\mu^2}\right)+\frac{\lambda}{8}u^2\left(1-\frac{\lambda}{16\pi^2}\ln\frac{\Lambda^2}{\mu^2}\right)\simeq \tag{F.15}$$

$$\simeq -\frac{\mu^2}{2}u\left(1-\frac{\lambda}{32\pi^2}\ln\frac{\Lambda^2}{\mu^2}\right)+\frac{\lambda}{8}u^2\left(1-\frac{\lambda}{32\pi^2}\ln\frac{\Lambda^2}{\mu^2}\right)^2=-\frac{\mu^2}{2}u_{\mathrm{R}}+\frac{\lambda}{8}u_{\mathrm{R}}^2,$$

which has the same structure of eq. (4.15).

The minimum of (F.15) coincides with the lowest-order value $E_U^* = -\mu^4/2\lambda$.

● RIASSUNTO

Si dimostra l'esistenza di configurazioni dei campi di gauge la cui densità di energia è inferiore a quella del vuoto perturbativo di Yang-Mills di una quantità che diverge come Λ^4 (Λ è il taglio ultravioletto). A questa singolare situazione fisica, che impedisce l'uso della teoria perturbativa anche a piccolissime distanze, è stato dato il nome di « instabilità essenziale ».

Резюме не получено.